

2. The roots of the equation

$$2x^3 - 3x^2 + 12x + 7 = 0$$

are α , β and γ

Without solving the equation,

(a) write down the value of each of

$$\alpha + \beta + \gamma \quad \alpha\beta + \alpha\gamma + \beta\gamma \quad \alpha\beta\gamma \quad (1)$$

(b) Use the answers to part (a) to determine the value of

(i) $\frac{2}{\alpha} + \frac{2}{\beta} + \frac{2}{\gamma}$

(ii) $(\alpha - 1)(\beta - 1)(\gamma - 1)$

(iii) $\alpha^2 + \beta^2 + \gamma^2$ (7)



1. The cubic equation

$$x^3 - 7x^2 - 12x + 6 = 0$$

has roots α , β and γ .

Without solving the equation, determine a cubic equation whose roots are $(\alpha + 2)$, $(\beta + 2)$ and $(\gamma + 2)$, giving your answer in the form $w^3 + pw^2 + qw + r = 0$, where p , q and r are integers to be found.

(5)

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6. The cubic equation

$$4x^3 + px^2 - 14x + q = 0$$

where p and q are real positive constants, has roots α , β and γ

Given that $\alpha^2 + \beta^2 + \gamma^2 = 16$

(a) show that $p = 12$

(3)

Given that $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{14}{3}$

(b) determine the value of q

(3)

Without solving the cubic equation,

(c) determine the value of $(\alpha - 1)(\beta - 1)(\gamma - 1)$

(4)

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3. The cubic equation

$$ax^3 + bx^2 - 19x - b = 0$$

where a and b are constants, has roots α , β and γ

The cubic equation

$$w^3 - 9w^2 - 97w + c = 0$$

where c is a constant, has roots $(4\alpha - 1)$, $(4\beta - 1)$ and $(4\gamma - 1)$

Without solving either cubic equation, determine the value of a , the value of b and the value of c .

(6)

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2. The roots of the equation

$$x^3 - 2x^2 + 4x - 5 = 0$$

are p , q and r .

Without solving the equation, find the value of

(i) $\frac{2}{p} + \frac{2}{q} + \frac{2}{r}$

(ii) $(p - 4)(q - 4)(r - 4)$

(iii) $p^3 + q^3 + r^3$

(8)

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3.

$$f(x) = \arcsin x \quad -1 \leq x \leq 1$$

- (a) Determine the first two non-zero terms, in ascending powers of x , of the Maclaurin series for $f(x)$, giving each coefficient in its simplest form.

(4)

- (b) Substitute $x = \frac{1}{2}$ into the answer to part (a) and hence find an approximate value for π

Give your answer in the form $\frac{p}{q}$ where p and q are integers to be determined.

(2)

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4. Use the method of differences to show that

$$\sum_{r=1}^n \frac{2}{(r+4)(r+6)} = \frac{n(an+b)}{30(n+5)(n+6)}$$

where a and b are integers to be determined.

(6)



4. Prove that, for $n \in \mathbb{Z}$, $n \geq 0$

$$\sum_{r=0}^n \frac{1}{(r+1)(r+2)(r+3)} = \frac{(n+a)(n+b)}{c(n+2)(n+3)}$$

where a , b and c are integers to be found.

(5)

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4. (a) Use the method of differences to prove that for $n > 2$

$$\sum_{r=2}^n \ln\left(\frac{r+1}{r-1}\right) \equiv \ln\left(\frac{n(n+1)}{2}\right)$$

(4)

- (b) Hence find the exact value of

$$\sum_{r=51}^{100} \ln\left(\frac{r+1}{r-1}\right)^{35}$$

Give your answer in the form $a \ln\left(\frac{b}{c}\right)$ where a , b and c are integers to be determined.

(3)

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2.

$$f(x) = \tanh^{-1}\left(\frac{3-x}{6+x}\right) \quad |x| < \frac{3}{2}$$

(a) Show that

$$f'(x) = -\frac{1}{2x+3} \quad (4)$$

(b) Hence determine $f''(x)$ (1)(c) Hence show that the Maclaurin series for $f(x)$, up to and including the term in x^2 , is

$$\ln p + qx + rx^2$$

where p , q and r are constants to be determined. (3)



2. (a) Write down the Maclaurin series of e^x , in ascending power of x , up to and including the term in x^3

(1)

- (b) Hence, without differentiating, determine the Maclaurin series of

$$e^{(e^x - 1)}$$

in ascending powers of x , up to and including the term in x^3 , giving each coefficient in simplest form.

(5)

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7.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Explain why, for $n \in \mathbb{N}$

$$\sum_{r=1}^{2n} (-1)^r f(r) = \sum_{r=1}^n (f(2r) - f(2r-1))$$

for any function $f(r)$.

(2)

(b) Use the standard summation formulae to show that, for $n \in \mathbb{N}$

$$\sum_{r=1}^{2n} r((-1)^r + 2r)^2 = n(2n+1)(8n^2 + 4n + 5)$$

(6)

(c) Hence evaluate

$$\sum_{r=14}^{50} r((-1)^r + 2r)^2$$

(4)

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